

Denise Holt: “The Potential Impact of Decentralized AI through Active Inference and Spatial Web”

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okay hi welcome back welcome back hi Denise hi thank you for for joining and looking forward to your presentation so take it away oh thank you so much and thank you for having me today um today I am going to talk about the potential impact of decentralized AI through active inference and spatial web Technologies and what we're really going to be discussing is um the potential aspects of this next era of computing uh it'll be incorporating a lot of the research that you're hearing about here over the next three days um and decentral in my talk decentralized AI is kind of taking a main focal point here and there's a couple of aspects um in technologies that are evolving right now that uh I think it's really important to be aware of um and they factor into uh what I'm going to be talking about today so one of the main uh things that is occurring right now is our internet is evolving um we have a new protocol layer that's layering right on top of the other protocols that are part of the same internet that we're talking across today um the difference is is what these protocols hstp hyperspace transaction protocol and hyperspace modeling language is the programming language um what these are going to enable is for us to have programmable spaces so instead of just programming pages in domains with content we are going to be able to program digital twin spaces of anything and this is going to enable context to be baked into everything in every space so all of the attributes about various things in various spaces and nested ecosystems and all of their inter relationships to each other so one of the things this is going to enable is uh active inference agents to be distributed across our internet and I see this as the next wave of innovation and through the through active inference these agents will be aware of each other and the network uh they'll be able to sense the world in real time and learn from each other and this is going to provide unprecedented levels of interoperability um so what we're talking about here is first principles uh automated systems and distributed collective intelligence so first let's look at the challenges in traditional centralized AI structures um or architectures uh centralized processing constraints you have high energy demands and scalability issues limited

adaptability to real time changes and you know we're all familiar with the limitations of uh these deep learning models they require massive amounts of labeled data and processing power they have limited ability to generalize across different tasks and environments AI models uh lack adaptability to real world complexity they're only good for one thing at a time uh like one uh optimized for one type of an output and they lack interpretability and transparency in decision making they have high computational resources leading to high costs and inefficiency and they struggle with generalization across different scales and tasks now we need decentralized Ai and decentralization really matters because it increases resilience and scalability for autonomous systems it reduces Reliance on Central servers and data hubs and it enables autonomy across distributed Networks so this Bridges the Gap in AI we move Beyond centralized AI for scalable autonomous systems that can operate independently and locally now we're all familiar with this guy here Dr Carl friston and uh he he gave a great talk uh first thing this morning on uh renormalizing General generative models um so what I'm going to be talking about are various components in in some emerging Technologies along with active inference that potentially realize this research paper designing ecosystems of intelligence from first principles now you know the this uh paper was published in December of 22 and um it really laid out a new framework for autonomous intelligence that mimics self-organized systems of nested intelligence found in nature and in in this paper it described a cyberphysical ecosystem of natural and synthetic sense making in which humans are integral participants what we called shared intelligence and this in the paper it says this V this vision is premised on active inference a formulation of adaptive behavior that can be read as a physics of intelligence and which inherits the physics of self-organization and in this context we understand intelligence as the capacity to accumulate evidence for a generative model of one's sensed World also known as self-evident of the abstract for this paper uh they also note we also consider the kinds of communication protocols that must be developed to enable such an ecosystem of intelligences and motivate the development of a shared hyperspatial modeling language and transaction protocol as a first and key steps towards such an ecology so what I'm proposing in this and what is proposed through all of this this uh technology together uh with all of the various parties that are involved in building these components is that the future of AI is shared distributed and multiscale an entirely new kind of autonomous intelligence able to overcome the limit s of machine learning Ai and as we know uh from understanding active inference this AI is knowable explainable and capable of human governance and it operates in a naturally efficient way there's no big data requirement and it's based on the same mechanics

as biological intelligence it learns in the same way humans so what I want to do is start with the spatial web protocol uh first because this is kind of the um the framework for these agents to be able to have this distributed intelligence structure so what is the spatial web protocol it's just the evolution of our internet into spatial domains um so people refer to it as web 3.0 it's just the next next evolution of the internet but this protocol has become a global standard a new Global standard for the last four years uh this this standard has been in development with the i e uh which is the largest core standards body uh in the world and back in 2020 they deemed the spatial web protocol a public imperative which is their highest designation and this summer it was voted and approved um as the new global standard and it's in the final administrative uh cycle with the i e before it becomes uh public so so there's a couple of things that are happening with this through hsm1 the programming language for this new protocol all emerging technology whether it's iot or ar VR distributed Ledger technology um you know all of these Technologies they're all going to be able to uh converge across the internet because they will have a common language for interoperability and in that you also have language that AI can understand and uh communicate together and communicate with these various aspects of Technology as well so this is going to produce the next era of computing and deoe called it back in 2020 so it's the next evolution of the internet and it's going to be empowered by active inference AI now just to kind of understand what this means for the internet um let's take a look at what a website domain is today with the worldwide web a domain basically contains a website and these websites they serve up content so whether it's you know images or video or audio or you know information data um that is what websites are capable of of doing now a spatial web domain is quite different because a spatial web domain could be anything in any space right and it can be programmable a spatial web domain is an entity with a persistent identity through time that has rights and credentials so you're talking about going from a library of pages in our Internet to the internet of everything and hsm1 becomes a common language for everything so it provides unprecedented levels of in operability between all current emerging and Legacy Technologies uh it's a programming language that Bridges communication between people places and things laying the foundation for a realtime World model for autonomous systems and what you have are programmable spaces hsm1 is a cipher for cont context anything inside of any space is uniquely identifiable and programmable within a digital twin of Earth producing a model for data normalization and from this we get adaptive intelligence automation security through Geo encoded governance multi- Network interoperability and it enables all Smart Technologies to function together within a unified

system and then this produces a Knowledge Graph and digital twins of anything everything potentially our planet so as this knowledge graph grows within the programming of all of these spaces across the network you get this Universal domain graph that enables a digital twin of our planet and all nested systems and entities within it we get nested ecosystems uh there there will be intelligent agents both human and synthetic across this network uh they will be sensing and perceiving continuously evolving environments making sense of the changes updating their internal model and what they know to be true at any given moment and acting on the new information that they receive so why do we need a spatial Internet Protocol well the increasingly graph-like nature of global data the opportunity for Automation and uh autonomic activities using context aware cognitive AI the need for composable systems and applications including the governance of such systems the intrinsic need for secure transactions the rise of machine learning and neural network computation and Edge Computing the need for explainable AI and robotic governance and the rise of iot and sensor mesh and some of the guiding principles of the spatial web standards are spatiality representing information as locations and relations in hyperspace ownership users own their data in digital property security secure data collection transmission and storage privacy individual control over digital identity and personal Data Trust reliable permission driven validation of users assets and spaces interoperability seamless navigation and asset transfer across spaces and responsibility creating technology ethically for the benefit of humanity and what has become of these Global standards is a socio technical standard so the spatial web is a socio technical system of systems uh takes into the account the um so the development of the spatial web must address both technological as well as sociological considerations socio technical standards integrate and balance the technical social physical and legal use of technology and the spatial web standard is a socio technical standard and what this results in is multiple scale cognitive Computing so the spatial web plays the role of a collective nervous system for smart infrastructure it hosts cognitive Computing uh inference perception learning and action selection by combining and abstracting distributed information the spatial web enables the formation of complex ideas and facilitates decision-making cognitive Computing occurs at multiple scales in the spatial web individual nodes host cognitive Computing functions and networks of nodes collectively perform emergent cognitive Computing so hsm1 hyperspace modeling language it's a human and machine readable modeling language and semantic data ontology schema there are some key elements to it there are entities activities agents contracts channels credentials domains and it includes hyperspace and time so hsm1

enables the expression and automated execution of Legal Financial and physical activities in the spatial web and all of these entities are identifiable and knowable by these intelligent agents through this uh language throughout the network now let's look for a second at the agent aspect agents and activities so agents and that can be human agent agents or synthetic agents um but these agents are entities that sense respond maintain a model of their environment and take actions to achieve goals and the activities that occur they're single actions or partially ordered sets of actions that are performed by the agents and they unfold over time so agents interact with domains and other agents to perform activities in the spatial web so what the spatial web facilitates is a distributed ecosystem of intelligence if you look at current state-of-the-art AIS they're siloed applications they're built for optimizing specific outcomes and they're they're really unable to communicate their knowledge frictionlessly and collaborate with other AIS but this facilitates a a an ecosystem of intelligence that is a self-evolving system system it's learning moment to moment and upgrading its World model and in that way it mimics biology while also enabling uh general intelligence so this is shared intelligence at the edge of everything so how active inference works in the spatial web it enables realtime decision making you have agents that are predicting and adapting to their environment from their frame of reference in the network uh they're continuously learning from their environment enabling realtime adaptation and unlike traditional AI which relies on pre-programmed rules and static models active inference continuously updates its models based on new data allowing it to adapt in real time so these active inference agents would seamlessly communicate in real time with each other sharing their knowledge and perspectives and learning from each other these active inference agents excel in handling complex uncertain environments by continuously minimizing uncertainty in fact they can quantify their level of uncertainty and this makes them suitable for applications where conditions are constantly changing so decentralized AI takes the processing to the edge to the edge devices you have a distribution of process in and power uh superpower gpus are not required the processing can take place on local devices throughout the network with infrastructure that already exists like your laptop or your mobile device or devices that haven't even been um invented yet um but we're already seeing a lot of those kinds of devices coming into play uh with you know um Apple Vision Pro and and you know the metal glasses and and different aspects of these Technologies so the processing takes place on these local devices throughout the network you have faster response times and reduced Network congestion and it uses real live data real-time data from iot sensors machines and everchanging context that's embedded

to the uh in the network so this is the sensory information that's coming in for these active inference agents so that they can uh you know do the perform this action perception feedback loop and continually update their model of the world of their reference and it eliminates the need to tether to a giant database active inference agents throughout the network learn and adapt from their own frame of reference within the network enhance data privacy by minimizing centralized storage and it minimizes complexity it uses the right data in the moment for the task at hand and with hstp with hyperspace transaction protocol you there's natural guard rails uh included in this framework so it creates a secure ethical and user Centric environment for AI applications fostering trust and reliability in digital interactions uh you have data privacy and sovereignty trans parency and explainability security and authentication bias mtig mitigation ethical AI guidelines interoperability standardization user empowerment and control accountability and governance safety and reliability Dynamic policy enforcement and what this results in is in intelligence at scale uh through this network of distributed intelligence it enables the cognitive architecture of the collective intelligence of multiples of agents that continuously communicate coordinate and collaborate with each other you have individual and specialized intelligences together on a common Network speaking a Common Language hsm1 efficient and Powerful cross communication to perform tasks regulate systems and address problems in real time and it scales up and grows in tandem with humans through hsm1 we can take human laws and guidelines make them programmable for these agents to understand and abide by in real time so we're talking about unprecedented levels of cooperation between human and machine intelligence and what you get then is an Ever evolving collective intelligence uh decentralization is an imper imperative for intellig at scale these agents represent scores of people from all over the world sharing their individual and specialized knowledge all coming together on a common Network and it's based on each agent's own world model you have unique perspectives from whatever frame of reference uh within the network nested ecosystems and levels of self-organization and Collective shared intelligence so the idea is that anybody can part their knowledge and their wisdom into these Agents from wherever they are in the world and this actually acts to preserve all of the uh cultural differences all of the um the belief differences the governing Styles um you know people whether they're in uh Kenya or Singapore or anywhere else in the world they should be able to create these agents to solve their local problems and then share that with with the world because then these agents can share their intelligence with each other so the spatial web protocol enables a decentralize secure and interoperable web with the potential to profoundly

transform Global Industries and everyday human life now let's dive into um kind of the the potential that exists for this idea of rgms these renormalizing generative models and the spatial web and we heard a great talk this morning uh by Dr friston he uh you know he talked a lot about the you know the science behind these uh these rgms so I'm going to kind of uh maybe maybe break down some of the aspects of how these work as leading up to then what they can do within this uh potential potentially with in this architecture so you know with rgms one of the interesting things is that it's a universal architecture um they're a UniFi unified scale-free framework for AI That's capable of reasoning learning and decision making and they can generalize across multiple tasks handling small and large scale data efficiently they combine perception learning and planning into a single system and they can be used to learn scalable architecture over space and time similar to human thinking they require significantly less training training data than traditional models while achieving Superior results so what are these rgms they're a new class of AI models based on active inference they use renormalization to simplify complex data into hierarchical structures and they operate on the free energy principle to minimize uncertainty and adapt in real time so the significance is that they represent a Leap Forward from traditional deep learning methods they reduce data requirements and uh improve efficiency and it's one single framework for all applications so the science behind rgms renormalization and multiscale learning renormalization is a technique from physics that's used to simplify complex systems to maintain consistency at different scales and rgms break down data into multiscale representations hierarchical structures rgms are actively learning from small details to large patterns simultaneously and they analyze like with an example would be like an rgm can analyze an image by recognizing edges then objects then scenes and understanding the the concepts at every at every scale so key Innovations and advantages of rgms why they're Superior to traditional models they learn efficient efficiently with less data it's a unified system that handles perception and planning simultaneously and they understand conceptual relationships not just patterns and they can adapt to new environments with minimal training uh aspects of them are scale-free modeling renormalization discrete representation hierarchical structures fast structure learning unified perception and planning and explicit handling of uncertainty so if you look at how the brain process data processes data um you know we we perform core screening all the time when we're learning right we break down sensory data into hierarchic patterns um that help us focus on the most relevant information um kind of this chunking of Concepts to to grasp different aspects of um whatever it is that we're we're learning or focusing on in the moment and rgms

can mimic this approach by processing data hierarchically focusing on important patterns so it's a first principles approach uh using a way to simplify models without losing important information um they can model space and time Dimensions like we do and can learn the causal structure of information and it helps handle large scale problems efficiently using less data and less energy now when you consider uh scale-free modeling you know um scale-free modeling these rgms they operate across different scales so from very fine details to highlevel con uh complex systems so unlike deep learning which struggles when moving between small and large scale problems this scale-free architecture allows rgms to seamlessly work on a variety of problems including small data problems and large scale strategic planning so when you when you're thinking of the concept of scale-free you know this is a great example here of a green apple you know you can you can see the green apple a concept of just you know pixels in an image or a volumetric object or on a branch on a tree or in an orchard and you know the the way that these uh rgms can model these conceptual understandings of things is it can understand all different scales of what a green apple is so when you think of like um the supply chain or or factory operations or or anything like that these agents can understand the what's happening across the manufacturing line on a very uh local level but also what's happening across the entire distribution line uh throughout the entire system so they can predict both short-term movements and long-term outcomes so you know the fine details as well as uh large larger uh outcomes with the highle complex systems all at the same time and they don't lose accuracy when switching between local and global data scales so hierarchical modeling in rgms hierarchical data processing rgms process at multiple levels of abstraction from fine grain details to highlevel Concepts uh they use recursive learning they continually Loop over data learning at each level continually refining their understanding of Concepts as they process new information and you get hierarchical representations that they build models of the World by zooming in and out of different layers of detail and this is similar to how human cognition processes both micro and macro scales so you know think about when you're when you're thinking about the concept of driving a car if you're talking to somebody about driving a car or you know you're you're focused on the aspect of driving you're not thinking about the car as all of its various parts that make up that complex system like the brakes and the steering wheel and the seats and the every aspect of the engine you just call it a car right because that's all you need to know in the moment for whatever whatever you're uh doing with that concept and so this is kind of a similar idea um and this layered approach allows rgms to understand complex multiscale problems without requiring vast data sets so conceptual

modeling for decision making um rgms then focus on conceptual modeling versus just mere pattern recog recognition um deep learning can recognize patterns but struggles with conceptual understanding it can recognize correlations without understanding relationships and it can recognize patterns but it doesn't understand the under lying meaning and rgms on the other hand focus on conceptual modeling they capture multiscale hierarchical structured Concepts that represent both fine grained and Abstract features it builds a hierarchical understanding of the world capturing cause and effect relationships so rgms understand the why behind patterns not just the what and the key difference rgms are not just a better generative AI but a transformative AI framework that builds world rgms can reason about cause and effect relationships making decisions based on conceptual understanding that are more aligned with real world outcomes so how do rgms leverage active inference by incorporating active inference Active Learning and active selection while incorporating realtime data to adapt its predictions and actions so what you get is realtime learning uh they adjust their beliefs based on new sensory input making decisions on the Fly optimizing both learning and decision-making processes and they also are able to uh have unified perception and planning so deep learning typically separates perception like recognizing an object from planning like deciding an action rgms can combine both in a unified model they handle both within the single model and learn Concepts that guide both recognition and future decisions so for example you can have an autonomous robot that can recognize a door handle and also plan how to open it using the same model so how what does all this have to do with the spatial web architecture so the potential that I see with rgms is that um the nature of how they work fits very well with the nature of this architecture within the spatial web and just to kind of bring you back to what I was talking about earlier you have hyperspace transaction protocol this manages data flow between interconnected systems and it facilitates interoperability between different AI systems and the hyperspace modeling language provides a common language for these systems enabling uh computable context and decisionmaking and it provides a shared framework for context aware decision-making so this protocol aims to create a standardized way for different Technologies and AI systems to interact and share information across programmable spaces within our global internet and this unified model this becomes a unified model for diverse applications handling complex multiscale systems through unified perception and planning it's a single framework that can handle multimodal tasks so you have versatility they can perform across multiple applications whether it's object recognition natural language processing strategic planning uni using a single Universal model and it reduces the

complexity uh it integrates the data analysis the ception and the strategy execution which is the planning within one model so you have cost savings it reduces the need for multiple specialized AI models it streamlines system management and saves resources so this how the spatial web enhances the rgm capabilities the combination of this scale-free active inference with the spatial web protocol could enable seamless integration of phys of the physical and digital these protocols allow rgms to operate across distributed networks exchanging insights and information between physical and digital environments enabling more sophisticated iot uh applications uh context aware Computing through the spatial web rgms would operate in context aware environments these agents uh adjust their behavior based on the phys physical and digital surroundings and interoperability multiple rgms these active inference agents and autonomous systems can operate in different environments whether it's your Home Smart City businesses and they could communicate collaborate and coordinate actions through the hyperspace transaction protocol there uh energy efficient distributed intelligence instead of relying on centralized energy intensive data centers intelligence could be distributed across networks of smaller more efficient devices and this this creates a nested ecosystem uh similar to a hallon architecture and if you're not familiar with a what a hallon is it's something that can be both a whole in and of itself and also part of a larger system so the spatial web and rgm would form a nested ecosystem where each node in the network whether it's a domain device agent or subsystem functions as a hulon it's autonomous yet interconnected contributing to a larger system's goals as part of the larger interconnected Network like a smart City infrastructure um your smart home could contribute the contribute data to your neighborhood grid which informs the city infrastructure decision all while functioning independently um if you if you look at this picture here of you know uh part of a supply chain image you know each item packed inside of each of those boxes is a hole in and of itself but it's subject to and it's part of the larger system the Box the box is subject to the the trucks that are moving it and distributing it or you know whatever's happening within the factory with those boxes and then that factory itself is its own hole that is subject to a larger ecosystem of the entire supply chain as a whole so when you're looking at what this next era of computing really enables it's just nested ecosystems that can be empowered by this natural intelligence and active inference Ai and the spatial web protocol can work together to create sophisticated adaptive systems at various scales so rgms for autonomy and self-organization with recursivity the scale-free models naturally form a nested hierarchy each level of abstraction in the model is complete in and of itself and yet part of a larger whole uh autonomy and cooperation active inference

agents exhibit both autonomy in their decision-making and the ability to cooperate with other agents and self-organization the renormalization technique introduced in the paper allows the system to dynamically organize information at different levels of conceptualization similar to the self-organizing nature of uh honic systems or these in these nested ecosystems so this last part um I want to explore the transformative potential of decentralized AI for the future of intelligence systems so the impact of distributed Collective intelligence the impact on this next era of computing would be enhanced problem solving and Innovation increased efficiency and scalability democratization of knowledge and Technology resilience and robustness enhanced decision making personalized and context aware Services ethical and responsible AI collaborative platforms and ecosystems enhanced learning and education empowered communities accelerated scientific research and environmental environmental and sustainability Solutions and the implications for future systems and Society the convergence of rgms in the spatial web could transform Industries enabling decentralized context aware systems that operate with minimal human intervention industries that could be impacted smart cities you would have autonomous systems that improve efficiency at all levels Global Supply chains rgms could optimize processes from local shipping to Global networks personal to critical systems homeostasis so whether it's personal devices or critical infrastructure a Continuum of intelligent adaptive systems can maintain homeostasis at each level in our Health Care Systems personalized treatment by integrating real-time data from wearables and medical records education we can have Aid driven adaptive learning systems that evolve with student needs and Environmental Management we can have ecological systems that could be modeled and managed more effectively from Individual habitats to Global Systems balancing local interventions with global goals and the future of AI and human civilization the development of renormalizing generative models within the scope of active inference and the spatial web promises to have farreach far-reaching implications more adaptive AI these systems are highly adaptable capable of operating in new environments with minimal training efficient Edge Computing rgms enhance the power of AI on devices like smartphones and iot sensors reducing the need for large data centers improved human AI interaction with explicit uncertainty models these systems interact naturally with humans seeking clarifications as needed safer AI systems they recognize and handle novel or ambiguous ambiguous situations reliably preventing catastrophic failures in unfamiliar scenarios so they can fail gracefully and accelerated scientific discovery rgms expedite research in field Fields like genomics and climate science by rapidly identifying patterns and formulating plausible hypotheses thereby

guiding further experimentation so the key takeaway here is that uh all of the all of this research and all of these Technologies together can provide us with a future where AI systems can interact seamlessly with the physical world autonomously optimizing operations and improving human lives so um yeah there's my website I have a podcast and I'll stop sharing now thank you Denise awesome I don't know what time how how's how are we on we're 15 minutes and I'll read some of the questions in the live chat okay sure I'll start with Pablo who wrote how will all of this affect the gamification of processes in organization s well I mean the the interesting thing is is through hsm1 you know all of these emerging Technologies are are going to converge together and become interoperable in ways that we've not experienced before you know we've had this um kind of vision of of this augmented reality existence but the framework hasn't been there and this is going to provide that framework so I think when you're talking about about gamification in every aspect it'll probably become kind of a natural part of of our existence and how we move through our day how we interact with with our work with uh brands with you know anything um you know I think that there's going to be a lot of of really interesting ways to kind of perpetuate that cool there's a lot of great questions in the chat so okay I'll read one from Tintin so they wrote it's impressive the obvious question I have is of alignment of this system in terms of that alien message from elaser gudowski for example and then they clarified what that exactly is so the key idea there is that simulated agents evolve and came to think faster than their creators so at a moment they took control to ensure their own survivability it's the question of how to ensure that way more intelligent and faster systems keep aligned with our values right and so one of the things that uh is really interesting to me about this is through hsm1 we can actually collaborate with these systems in the sense that we can take human laws and guidelines and make them programmable so these agents can understand and abide in real time and you know um versus uh who you know Dr Carl friston is is their cheap science scientists um they were involved in a program called Flying forward 2020 and and they were doing a lot of uh testing around this and because it was it involved a drone project that I I don't know there were multiple countries throughout Europe it was with the European Union that were involved and you know testing whether the they could program in laws around airspace laws and you know things like that they did all kinds of scenarios of like you know Medical Supply delivery uh perimeter security all kinds of things and what they found was through hsm1 you could actually uh program these guidelines and these these agents abide by them so I think we are going to experience this unprecedented level of of cooperation and collaboration so we can grow in tandem with the you know with

these uh agents and their intelligence and I think that we can also you know take some um solace in the fact of how you know active inference works and you know how our human knowledge grows you know I think we're going to see a similar pattern with this knowledge among these agents and the only difference is and and this is just me speaking from you know my own you know my own uh perception and and perspective of this but you know they're going they're going to be dealing with real-time data and with real uh a real grounding of this programmed these programmed spaces and those are the things they're going to be operating off of to make their decisions right whereas if you look at a human you know we have so many other variables going on and every when we learn through active inference it's a very subjective uh growth that's why you can have you know multiple children growing up in the same house household and they all experience things differently right but when you have these agents and they're getting this they're getting their learning through actual tangible programmed information I think there's a a stability there as well if that makes sense and and and again that's just my own uh you know interpretation and feeling around this and I could be I could be wrong but you know just kind of throwing that out there that's how we move forward yeah okay next question from David Highland who wrote thank you for the informative presentation Denise can you share any thoughts or information about the ownership model for agents is it a decentralized market how do we decide which parts are public goods yeah so with the protocol with a with hyperspace transaction protocol um because so it's interesting it really shifts to this uh kind of self- sovereign identity uh aspect of the internet because right now think about how web domains work you know if you want to interact with a web domain all the transactions are taking place under the umbrella of whatever centralized organization owns that domain and you have two choices you can either say yes you can have all my data and you can decide what to do with it all or I don't want you to but then I'm opting out of being able to engage with your domain well in the spatial web because now transactions can be at every touch point of every entity within the network then permissions can be programmed in a very nuanced way right so it really puts that back in the hands of the of the user of of the entity within it you can you can program you can program guardrails around certain aspects of data while allowing permissions and Freer like Freer accessibility for other aspects of data so whether you're talking about a business you know they can guardrail off their data they can guardrail off certain aspects of it for certain Departments of employees some for their customer base some for you know um Partnerships whatever right um and we're going to be able to do that kind of thing too and the thing with

this is this takes into account time the passing of time so you can set expirations on permissions as well so I think we're going to have a lot more self- Sovereign control in this in this area yeah that's a part that really excites and Peaks me as well like with the web domain or the email domain it'd be like saying well on one hand we're not going to look in your email let's just go with that for now but we do or don't accept letters from this area so it's kind of all or not nothing and it's based upon like you pointed out the the custodianship of the all or none permission based around what boxes you can send where whereas having a more nuanced way to talk directly about belief sharing means that instead of just like plugging the USB into the port there could be even a active inference negotiation around what elements to share how to share when to expire all these other features which can can help like incrementally build rapport and and add to the relationship instead of doing all or none which can be just very prone to false positive or false negative in cyberphysical interactions yeah you're so right Daniel yeah what do you think about learning trajectories and paths I know it's a interest of all of ours I guess like what what can we do to be learning and thinking about all these things well um so you know I I have I actually have put together a series of courses um to kind of give people a a real um kind of a prerequisite knowledge understanding of what this transition looks like and the different components and what it means from all aspects of like you know basic understanding of of active inference and uh you know the free energy principle and the spatial web Technologies to you know decentralized AI what does that mean convergence of all this Tech across the network what do that look like safe AI how does active inference benefit that and then rgms let's let's look a little closer at at that and what I try to do is I try to take the complex ideas but make them more understand so that people can really wrap their heads around the these Concepts and then they can go dive into the more technical aspects of it and I know you with the Institute you have a lot of technical knowledge there so I think that you know between resources like you know some of my resources your resources and then I'm sure there are resources out there that maybe I'm not even aware of and you know I think there's going to be more and more resources available around this modeling these agents you know uh building out you know these these spaces uh within the network um I know that versus is going to be launching a platform that'll be the first interface for the public around this technology and I think it's coming to the public at the end of first quarter next year and with that there will be tools where people will be able to build these agents as applications with in the network and and you know I think at that point it's going to get really exciting um so yeah I I just think over the next six months we're going to see a lot coming about with this and if

anybody is interested in the courses I've put together you know you can visit my uh my website and and sign up for them uh they're not available yet but they'll be available very shortly uh within the next week or two so yeah cool okay I'll I'll for last question ask the same thing that I asked uh Alex and Carl earlier what is a fun or a memorable moment from learning and applying active inference this year and then if you haven't already mentioned it what something that you're excited for next year um this year honestly I I think that uh I have been on this Learning Journey in creating this course material around it you know and at the same time I've been building community and kind of having them be my test guinea pigs around it and stuff so I you know there's been this whole learning aspect for me around building this educational material so it it has been you know a constant uh feedback loop and next year I'm just very excited about the potential of um building these agents and what that means and and how we can tackle some of the the problems and the the issues and the challenges that we have um through this new exciting technology and this this uh framework so yeah well thank you for all of your great learnings and digested and and sharings and and all the ways that that what you talked about today I think connected a lot of the dots from hearing Carl's more technical view on the rgm on through what what does it mean to have multi-agent decentralized as opposed to just building One agent so thank you again Denise thank you Daniel thanks for having me thanks everybody till next time bye e